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PDA Research

2023 PDA Visual Inspection Survey

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2023 PDA Visual Inspection Survey

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Introduction

Overview

Visual inspection continues to be a critical step in the manufacture and release of high-quality injectable medicines. There is limited specific guidance on inspection methods or acceptance criteria which has led to a wide range of industry practices. This diversity is also to be expected given the variety of products and packages currently being manufactured and thus inspected. In recent years we have also seen increased interest and activity by the regulatory authorities in this part of the process, providing additional incentive for consistent practices. In 2023, the fifth survey in a series of surveys was completed by PDA to better understand current industry practices in this important area. Past surveys in 1996, 2003, 2008 and 2014 have provided practical guidance and insight to those working in this field. The purpose of this survey was to document current industry practice for visual inspection of injectable products.

The survey was sent directly to all PDA members and was open to nonmembers as well. For some questions, respondents were allowed to check multiple options, therefore the sum of the response percentage may exceed 100% and the total response count may exceed the number of respondents to the survey. Percentages shown for each question are based on the number of responses for that question. In some cases, raw responses for a specific survey question had to be combined to allow relevant comparisons to the historical survey results. In these cases, the survey question has a footnote indicating how the raw responses were combined. The most frequent response to a given question is highlighted in **red**.

The results presented for the 2023 survey are based on 187 responses received between November 3, 2022, and January 31, 2023. Each of a company's manufacturing sites could provide an answer; therefore some companies have more than one response. PDA conducts its benchmarking surveys in a manner designed to protect the confidentiality of the gathered information and data. The identity of survey respondents was blinded and not revealed to the authors or other PDA members, or in any publication or presentation of the final results developed on the survey. The same population was sampled for each survey year, but the specific companies and manufacturing sites that responded each year may be different. This limits to some degree the identification of trends. Remember, the survey documents current industry practice, but does not indicate if these are good or bad practices.

The numbering in this report does not coincide with the numbering of the questions in the actual survey. Preliminary results from this survey were originally presented at the PDA Visual Inspection Forum held April 17-18, 2023 in Baltimore, MD.

Summary of Results

This survey represents the global pharmaceutical industry with a good distribution of sites worldwide. Most responses came from North America (39%), Europe (36%) and the Asia/Pacific region (17%). Good geographic representation also extended to the markets supplied by these plants including North America (79%), Europe (68%), Asia/Pacific region (53%), Japan (42%), South America (43%) and Africa (35%). Small and large manufacturers were also well represented with small volume sites (<1M units/year) accounting for 24% of the respondents and large volume sites (>100M units/year) accounting for 19%. These results are similar to the previous survey year (2014) with an increase in European and a decrease in Japanese sites that responded.

The majority (86%) of surveyed products inspected are for human use and include a significant amount (44%) of biological/biotech products. A new question regarding cell/gene therapy or Advanced Therapy Medical Products (ATMP) was added to the survey for this year with a response of 29%. Most are aqueous solutions (86%) or lyophilized powders (49%), similar to the last survey year.

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These products are mostly packaged in molded (64%) and tubing (60%) glass vials, with a significant number in glass syringes (42%) and plastic vials (36%). This is a marked increase in the use of molded glass and plastic vials.

Manual inspection continues to be the most used method for both particles (50%) and the container/closure system (48%). There is continued interest in using automated inspection with 57% of firms having plans to implement such systems in the next two years. Similar results have been observed in previous surveys. Automated systems are validated with prepared standards (78%) and/or production defects (51%) to perform equivalent or better than manual inspection (77%).

Most firms (83%) control manual inspection time and do not use magnification. 47% of firms inspect more than a single container at a time. The median inspection time was 6-10 sec per container which agrees with the current European Pharmacopeia (EP), Japanese Pharmacopoeia (JP) and United States Pharmacopeia (USP) inspection conditions. Illumination intensity is typically 2,000-3,750 lux (71%) which also agrees with the current compendial inspection conditions with some (22%) using higher values.

Inspection continues to be performed most often (68%) offline, but a significant amount (52%) is also performed in-line with packaging and in-line with filling (48%). Training (100%), a test of visual acuity (100%) and inspection performance (94%) are part of the typical inspector qualification process. Annual requalification (87%) continues to be the typical time interval used for human inspectors.

Test sets with 100-300 units (30%) and 301-500 (28%) and with a defect rate of 5-10% (30%) are used most often for inspector qualification. Inspectors are given a 5 minute (34%) or 10 minute (32%) break every 60 minutes (46%) or every 30 minutes (26%). Most firms use the same inspection conditions for different regions (88%), veterinary products (89%) and/or clinical supplies (96%).

The typical total reject rate is 1-2% for aqueous solutions and <1% for lyophilized powders. Differences in typical reject rates are likely due to detection ability rather than differences in the underlying particle load. Particles, and specifically lint/fibers, continue to be the most common defects observed.

After 100% inspection, lots are routinely (84%) audited on a per batch basis, most often (71%) by QA per equivalent standards ANSI/ASQ Z1.4 or ISO 2859. The median values for AQL's used with these plans are 0.065% for Critical, 0.65% for Major and 2.5% and 4.0% for Minor defects. A new question was added asking the AQL used for particles and the median response was 0.65%, which agrees with the acceptance criteria found in the USP. The high number of respondents who reported zero for the Critical defect AQL likely suggests a high use of "none allowed" for Critical defects as expected but also a continued misunderstanding of the difference between AQL values and accept numbers. Because the AQL value is based on a sample taken from a larger lot, the AQL value will always be greater than zero even with no defects (accept number of 0) accepted in the sample. The median AQL values for Critical and Major defects have not changed from the 2014 survey. The median AQL value for Minor defects was equally divided between 2.5% and 4.0%.

Conclusions

Inspection continues to be performed with a variety of methods. The rapid increase in the use of automated inspection for container/closure systems in previous years was likely driven by technological development. In the current survey year, the choice to use manual, semiautomated or automated inspection technology for particle and/or container/closure inspection is likely a function of the volume of product to be inspected rather than significant differences in inspection performance. Specifically, larger volumes support the significant capital cost and validation expense of automated systems.

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Smaller volumes and more diverse product mix likely leads to manual inspection with semiautomated inspection filling the space between.

The most commonly used test method (67%) for Difficult to Inspect Products (DIP) is reconstitution of lyophilized powders as part of a supplemental inspection as discussed in USP <790>, USP <1790> and PDA TR79. Many firms have established alert/action limits based on 100% inspection results (92%) and investigate (91%) and/or reinspect (68%) when these limits are exceeded.

The publication of USP <790> and <1790> continues to move firms to align with the manual inspection conditions specified therein and also found in the EP 2.9.20 *Particulate Contamination: Visible Particles* and more recently in the JP 6.06 *Foreign Insoluble Matter*.

Finally, 40% of firms have been challenged by a regulatory inspector on their inspection method or acceptance criteria in the last two years. 31% of firms expect changes in customer expectations in the next five years. 31% of firms expect changes in customer expectations in the next five year; specifically, a tightening of visible particle limits. 44% of firms expect changes in regulatory expectations in the next five years as well; again, with a tightening of visible particle limits. These are both lower than in the previous survey years and may indicate some degree of stabilization of VI expectations.

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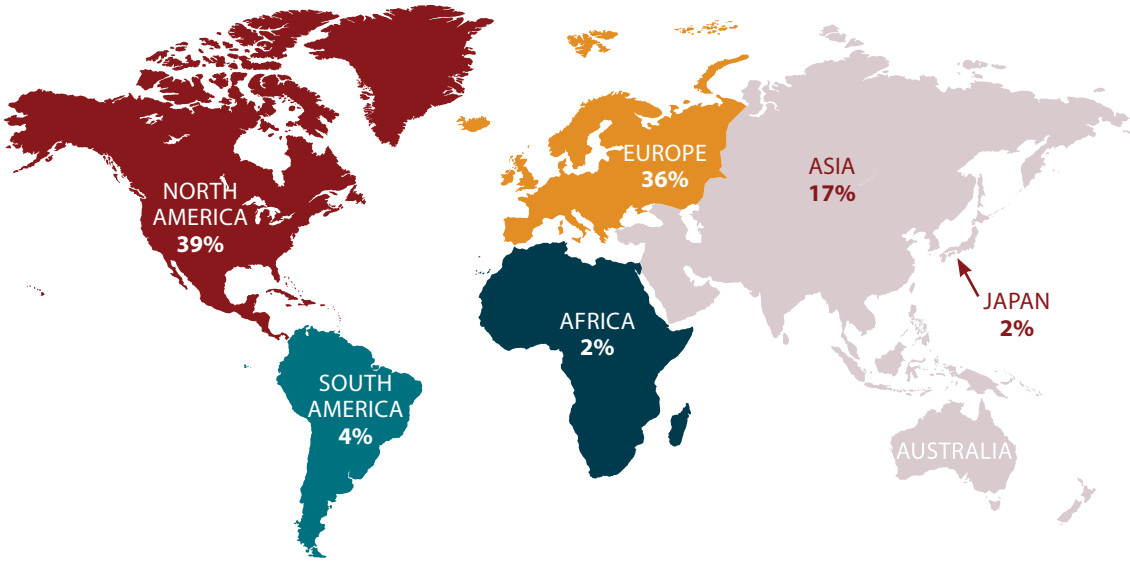
General Information (Questions 1.1- 1.10)

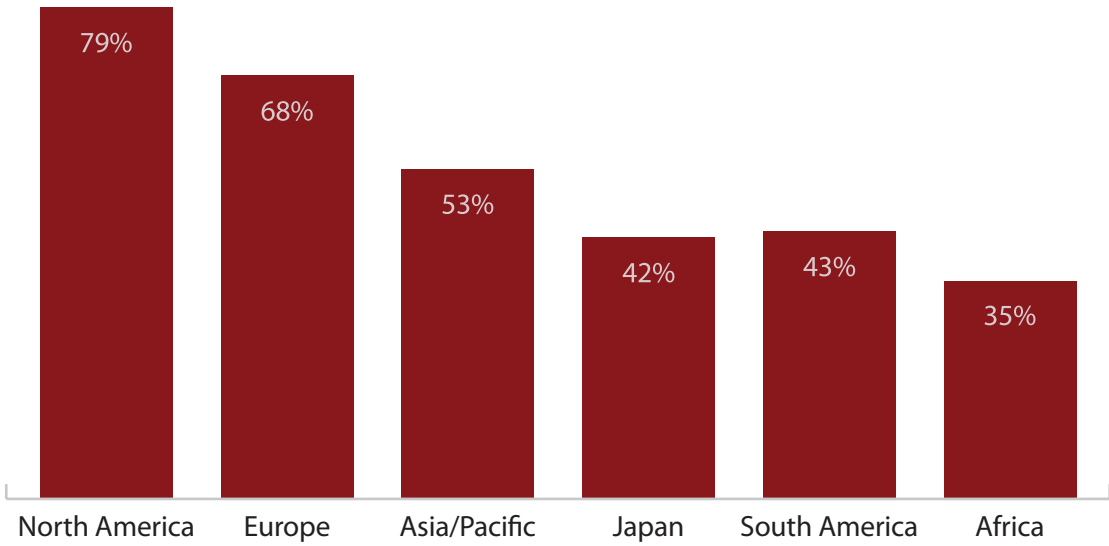
1.1 In what geographic region is this facility located?

Total respondents: 183



1.2 To what geographic regions are products manufactured at this facility distributed?

Total respondents: 184



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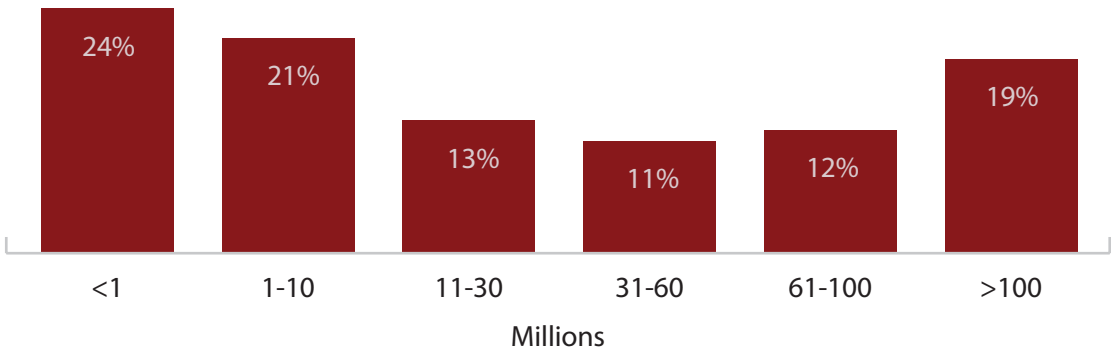
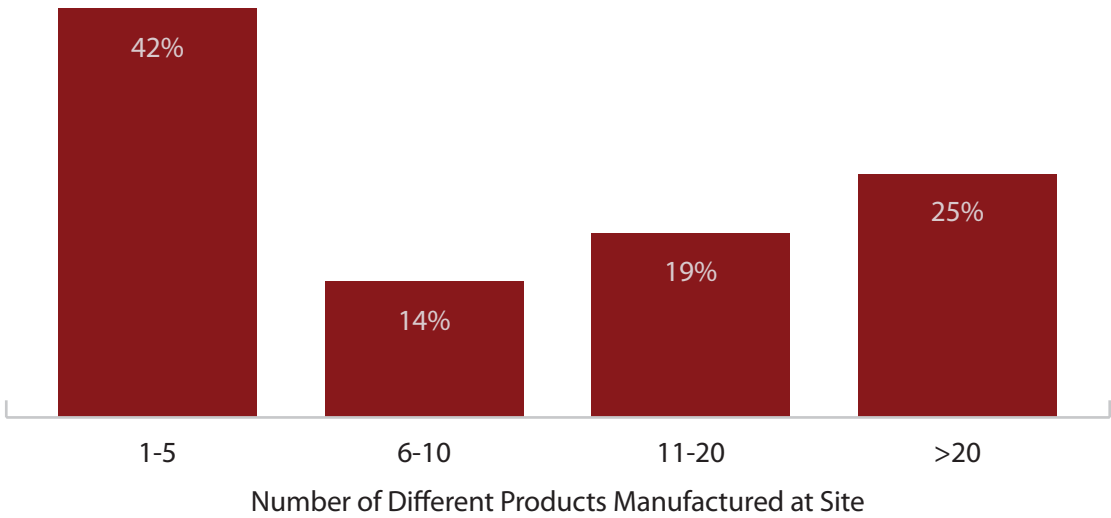
1.3 What is the approximate total number of injectable units produced at this facility?

Total respondents: 182



1.4 How many different injectable products are produced at this facility?

Total respondents: 184



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1.5 How would you best describe your company?

Total respondents: 184

	2023	2014	2008	2003	1996
Innovator Manufacturer	**49%**	ND	ND	ND	ND
Generic or Biosimilar Manufacturer	**30%**	ND	ND	ND	ND
Contract Manufacturing Organization (CMO)	**21%**	ND	ND	ND	ND

ND = No Data, question not asked in survey from this year

1.6 What types of products are manufactured at this facility?

Total respondents: 183

	2023[1]	2014	2008	2003	1996
Human Health	**86%**	77%	67%	85%	80%
Biological/Biotech	**44%**	54%	76%	37%	40%
Cell or Gene Therapy / ATMP	**29%**	ND	ND	ND	ND
Medical Device / Combination Product	**29%**	22%	ND	ND	ND
Animal Health	**25%**	14%	48%	7%	10%
Diagnostic	**20%**	15%	5%	4%	10%
Other[2]	**11%**	ND	ND	ND	ND

ND = No Data, question not asked in survey from this year

[1] Quartile survey results transformed to a single value by summing responses across quartiles.

[2] Other includes topical ophthalmics, nasal sprays, creams, and implants.

1.7 What are the product formulations produced at this facility?

Total respondents: 183

	2023[1]	2014	2008	2003	1996
Aqueous Solutions	**86%**	84%	54%	40%	60%
Lyophilized	**49%**	59%	25%	30%	27%
Suspension	**42%**	34%	6%	22%	9%
Oils and Emulsions	**26%**	10%	9%	3%	1%
Powder	**19%**	17%	0%	1%	2%
Other[2]	**9%**	ND	ND	ND	ND

ND = No Data, question not asked in survey from this year

[1] Quartile survey results transformed to a single value by summing responses across quartiles.

[2] Other includes nonaqueous solutions, implants, blood products.

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1.8 What common primary containers are used at this site?

Total respondents: 173

	2023[1]	2014	2008	2003	1996
Molded Glass Vial	**64%**	55%	15%	19%	35%
Tubing Glass Vial	**60%**	70%	42%	48%	55%
Glass Syringe	**42%**	40%	11%	0%	4%
Plastic Vial	**36%**	14%	2%	0%	0%
Flexible Bags	**31%**	6%	ND	ND	ND
Cartridges	**29%**	20%	ND	ND	ND
Blow-Fill-Seal	**22%**	5%	ND	ND	ND
Glass Ampoule	**24%**	29%	15%	7%	1%
Plastic Syringe	**28%**	15%	1%	0%	1%

ND = No Data, question not asked in survey from this year

[1] Quartile survey results transformed to a single value by summing responses across quartiles.

1.9 Where do you perform 100% inspection?

Total respondents: 124

	2023[1]	2014	2008	2003	1996
Offline	**68%**	79%	81%	59%	37%
In-line with Packaging	**52%**	58%	3%	17%	42%
In-line with Filling	**48%**	43%	16%	22%	31%

[1] Quartile survey results transformed to a single value by summing responses across quartiles.

1.10 What technique is used most often for inspection for particles or inspection of the container/closure?

Total respondents: 149

	2023[1]	2014	2008	2003	1996
Particles:					
Manual	**50%**	49%	33%	46%	33%
Semiautomated	**21%**	17%	24%	19%	20%
Automated	**30%**	33%	43%	35%	42%
Container/Closure:					
Manual	**48%**	54%	36%	63%	48%
Semiautomated	**20%**	18%	26%	15%	42%
Automated	**29%**	28%	39%	20%	5%

[1] Quartile results transformed to a single value using a weighted average.

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Manual Inspection (Questions 2.1- 2.13)

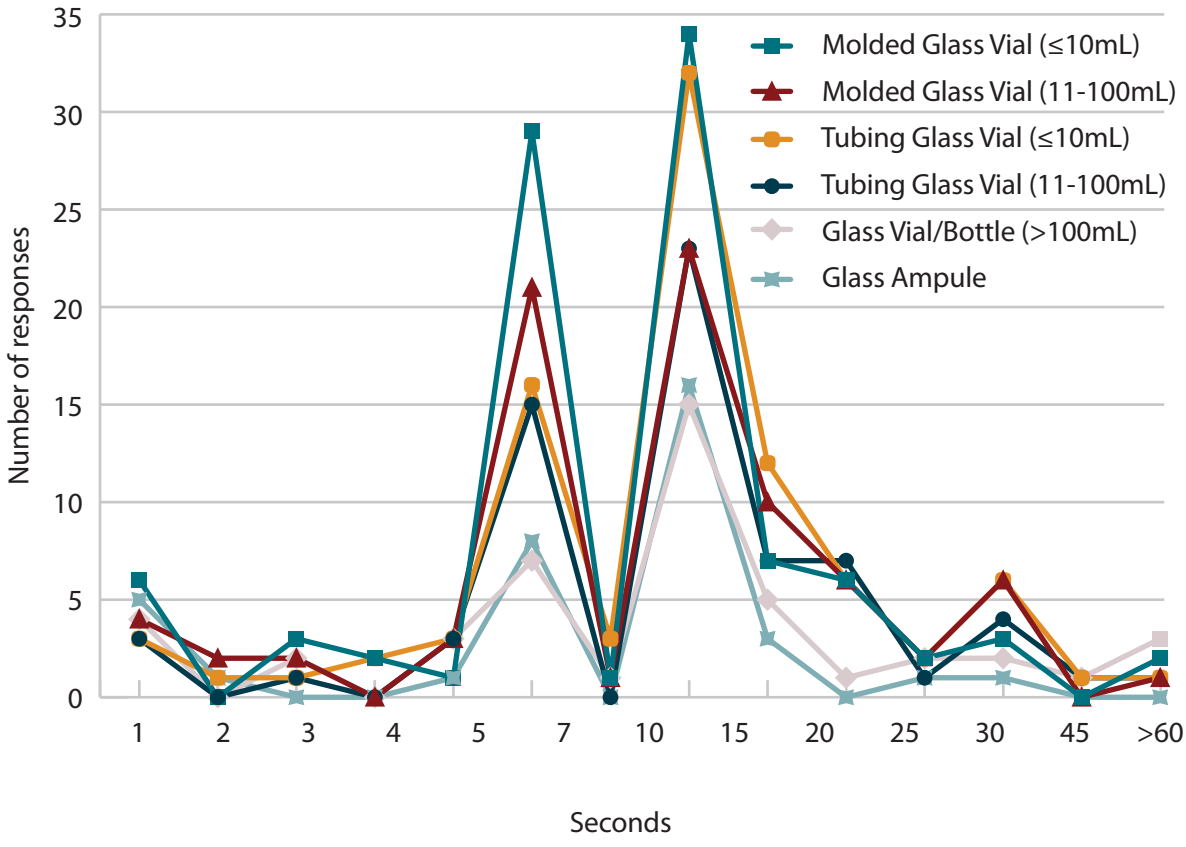
2.1 Summary of responses regarding manual inspection conditions.

Total respondents: 153

- 83% control inspection time or the pace of inspection.
  - **80% by SOP**
  - 33% with Timer
  - 19% with Conveyor
- 15% use a magnifier.
  - **29% 2X**, 24% 3X, 10% 4X, 24% 5X, 14% >5X
- Light Source used:
  - **62% LED**, 44% Fluorescent, 12% Incandescent

2.2 What is the average inspection time for this container type?

Total respondents: 131



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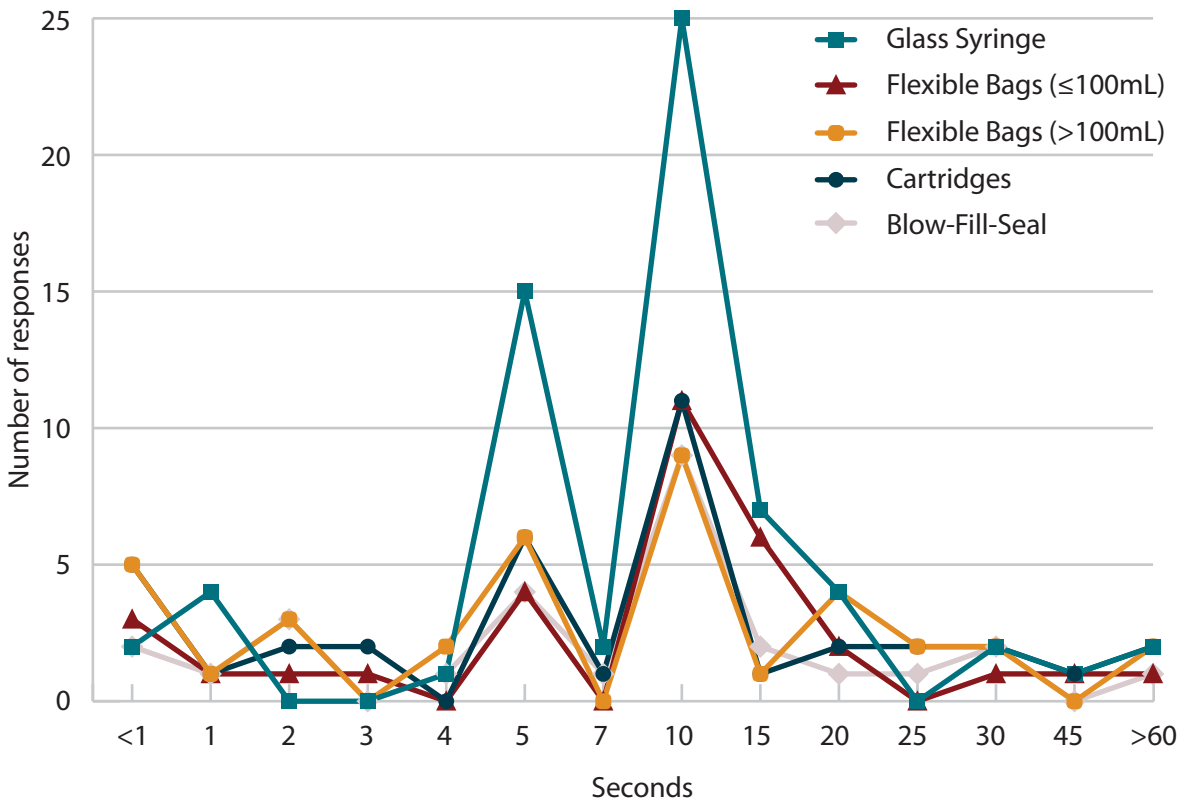
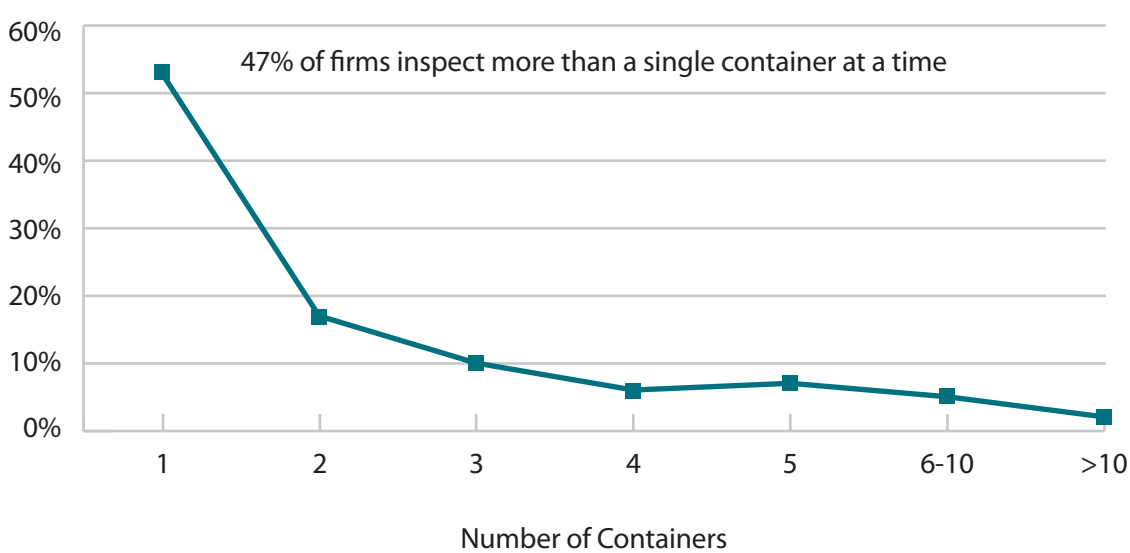
2.3 What is the average inspection time for this container type?

Total respondents: 105



2.4 On average how many containers are inspected together each time (e.g., by hand, with a clip or other device)?

Total respondents: 175



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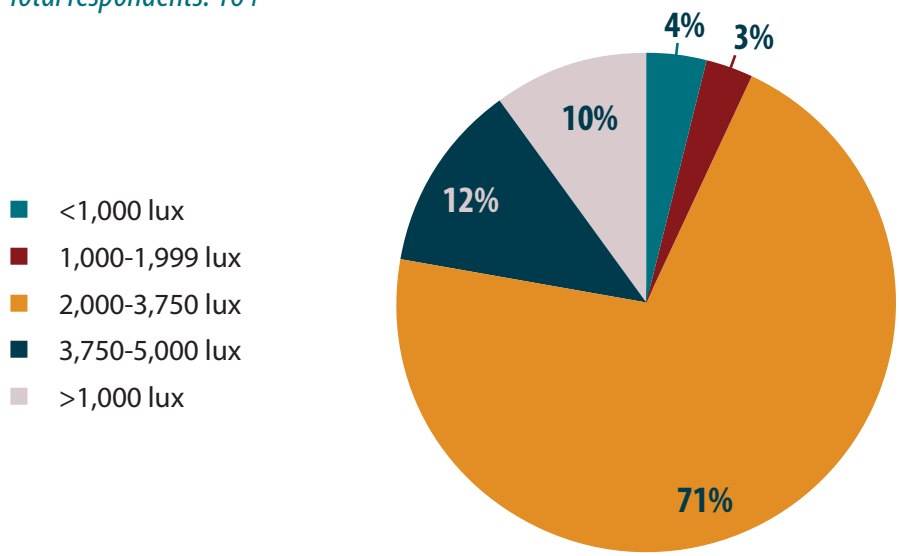
2.5 What is the average illumination intensity at the container during manual inspection?

Total respondents: 164



2.6 Inspector Selection Criteria

Total respondents: 169

	2023	2014	2008	2003	1996
Training	**100%**	94%	89%	96%	80%
Visual Acuity	**100%**	91%	79%	85%	80%
Test of Inspection Ability	**94%**	90%	100%	89%	80%
Color Vision	**86%**	74%	68%	ND	ND
Education	**ND**	41%	26%	30%	25%
Experience	**45%**	32%	37%	15%	30%

ND = No Data, question not asked in survey from this year

2.7 How are inspectors qualified?

Total respondents: 172

- 99% describe defects and inspection conditions in a written procedure.
- The qualification conditions are:
  - **Simulated/Offline: 84%**
  - Actual Manufacturing: 33%
  - Worst-case fatigue: 63%
- 89% use test kits to qualify inspectors.
- These test kits are composed of:
  - **Production Defects: 76%**
  - Nonspherical Standards: 40%
  - Spherical Standards: 30%

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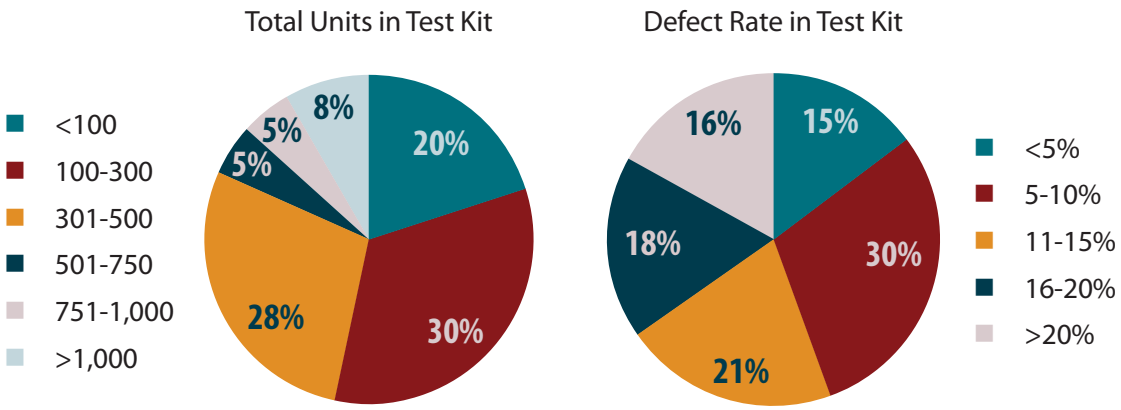
2.8 The composition of test kits used to qualify inspectors.

Total respondents: 147



2.9 How often are inspectors requalified?

Total respondents: 175

	2023	2014	2008	2003	1996
Never	**0%**	5%	21%	8%	35%
Monthly	**0%**	1%	5%	0%	8%
Quarterly	**1%**	4%	0%	0%	8%
Semiannually	**11%**	10%	11%	8%	16%
Annually	**87%**	79%	63%	75%	69%
Biennially	**1%**	ND	ND	ND	ND

ND = No Data, question not asked in survey from this year

2.10 78% of respondents indicated that they require requalification after an extended absence.

Total respondents: 172

- How long after an absence triggers requalification?
  - 1 month: 6%
  - 3 months: 46%
  - 6 months: 26%
  - >6 months: 23%

2.11 74% of respondents indicated that they apply the same training and qualification criteria to both Production and QA Inspectors.

Total respondents: 172

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2.12 How frequently do inspectors take a break or rotate to a noninspection task?

Total respondents: 157

	2023	2014	2008	2003	1996
Never	**4%**	2%	ND	ND	ND
<30 min	**4%**	2%	16%	12%	5%
30 min	**26%**	33%	32%	15%	21%
45 min	**6%**	3%	ND	ND	ND
60 min	**46%**	47%	32%	62%	32%
2 hrs	**12%**	9%	11%	12%	37%
4 hrs	**2%**	4%	0%	0%	5%

ND = No Data, question not asked in survey from this year

2.13 How long are these breaks?

Total respondents: 155

	2023	2014	2008	2003	1996
1-5 min	**34%**	50%	ND	ND	ND
6-10 min	**32%**	17%	ND	ND	ND
11-15 min	**17%**	20%	ND	ND	ND
>15 min	**17%**	13%	ND	ND	ND

ND = No Data, question not asked in survey from this year

Automated Inspection (Questions 3.1 – 3.6)

3.1 Does your firm have plans to replace manual inspection with automated inspection?

Total respondents: 172

	2023	2014	2008	2003	1996
Shift to Automated Inspection	**57%**	50%	67%	50%	68%
Justification:					
Quality	**66%**	85%	75%	92%	92%
Productivity	**ND**	87%	92%	92%	100%
Increased Capacity	**69%**	ND	ND	ND	ND
Cost Savings	**54%**	ND	ND	ND	ND
Ergonomics	**2%**	ND	ND	ND	ND

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3.2 If you validate your automated inspection equipment, what was the acceptance criterion?

Total respondents: 104

- **Equivalent or better than manual: 77%**
- Other, Not compared to manual: 19%
- Hybrid, Particles and/or Critical and Major defects equivalent or better than human and Minor defects compared to a fixed criterion: 4%

3.3 When performing qualification or validation do you do the following:

Total respondents: 138

- **Challenge sensitivity with known defects: 88%**
- Assess false reject rate (FRR): 63%
- Use tightened AQL to assess performance: 42%
- Assess performance w/ 100% inspection by both MVI and AVI: 37%
- Test three (3) commercial batches: 33%

3.4 How frequently do you challenge or retest automated inspection equipment?

Total respondents: 96

	2023	2014	2008	2003	1996
Never	**4%**	1%	0%	0%	15%
Each Shift	**7%**	1%	8%	13%	8%
Start of Lot	**50%**	46%	42%	75%	38%
End of Lot	**3%**	ND	ND	ND	ND
Start and End of Lot	**1%**	8%	ND	ND	ND
Daily	**15%**	15%	25%	19%	23%
Weekly	**2%**	2%	0%	0%	8%
Monthly	**0%**	2%	ND	ND	ND
Quarterly	**4%**	1%	ND	ND	ND
Annually	**25%**	19%	ND	ND	ND
Other[1]	**3%**	ND	ND	ND	ND

[1] At recipe change, after maintenance

ND = No Data, question not asked in survey from this year

3.5 What type of samples do you use for routine challenges?

Total respondents:115

- **Prepared Standards: 78%**
- Production Defects: 51%

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3.6 If you use automated inspection equipment, have you investigated or implemented a system using artificial intelligence (AI)?

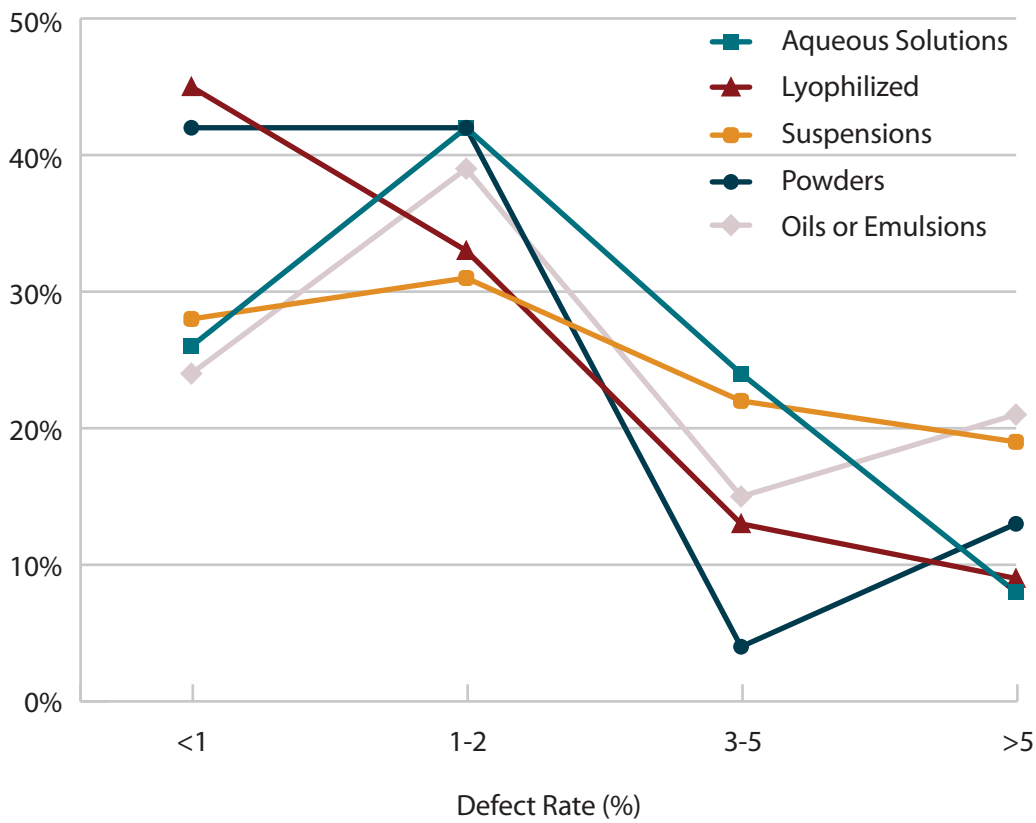
Total respondents: 99

- **No: 49%**
- Investigating: 44%
- Implemented: 7%

Inspection Results (Questions 4.1 – 4.3)

4.1 What is the average reject rate for this product formulation?

Total respondents: 146



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4.2 What are the most common defects found during visual inspection? (Rank order with 1 most frequent)

Total respondents: 142

	2023	2014	2008	2003	1996
Particles	**1**	1	1	1	1
Scratches	**2**	2	2	4	4
Crimp Seal	**3**	3	3	3	2
Cracks/Chips	**3**	4	5	2	3
Cap	**4**	5	6	7	9
Stopper/Plug	**5**	7	8	9	8
High/Low Fill	**6**	6	4	5	5
Cake	**7**	8	8	6	6
Leaks	**7**	9	7	8	7

4.3 What are the most common types of particles found during visual inspection? (Rank order with 1 most frequent.)

Total respondents: 139

	2023	2014	2008	2003	1996
Lint/Fiber	**1**	1	1	1	1
Product Related	**2**	3	3	4	3
Glass	**3**	2	2	2	2
Rubber/Elastomer	**4**	4	4	5	5
Metal	**5**	5	5	3	4

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Acceptance Sampling and Inspection Strategies (Questions 5.1 – 5.12)

5.1 Do you audit or perform a sampling inspection (AQL inspection) after 100% inspection and who performs the audit?

Total respondents: 145

	2023	2014	2008	2003	1996
Audit Frequency:					
Audit every lot	**84%**	92%	85%	72%	90%
Audit selected lots	**5%**	0%	0%	8%	5%
No audit	**10%**	8%	15%	20%	5%
Personnel performing the audit:					
Audit by QA	**71%**	71%	74%	85%	89%
Audit by Production	**29%**	29%	26%	15%	11%

5.2 What sampling plan does your facility use?

Total respondents: 112

	2023	2014	2008	2003	1996
ANSI/ASQ Z1.4	**63%**	65%	53%	70%	90%
ISO 2859	**29%**	23%	11%	10%	0%
JIS Z9015	**0%**	7%	15%	5%	0%
Mil Std 1916	**2%**	3%	11%	0%	0%
Dodge Romig	**1%**	1%	0%	5%	0%
Other[1]	**5%**	2%	10%	0%	10%

[1] 2×100% inspection, plan based on AQL and RQL

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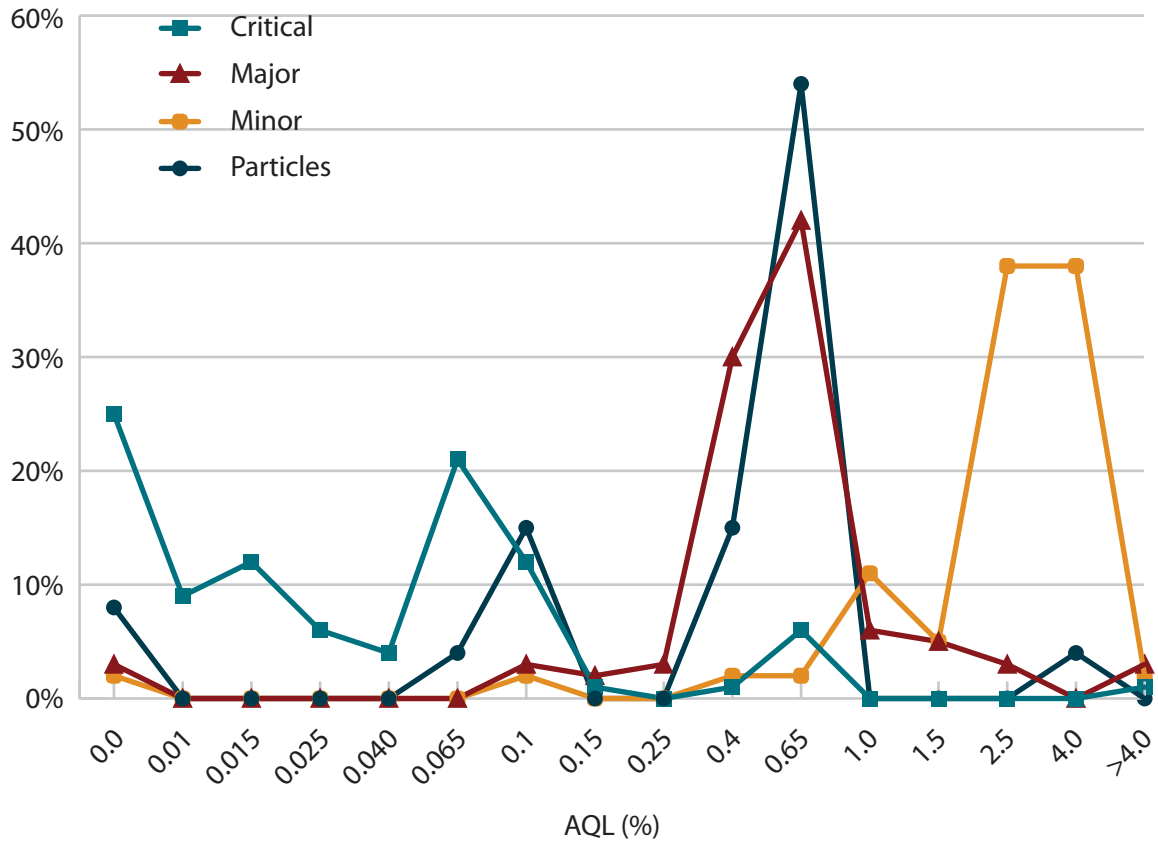
5.3a What AQL value (in %) do you use for acceptance sampling of these defect categories?

Total respondents: 69



5.3b Most frequently cited AQL value for each criticality:

	2023	2014	2008	2003	1996
Critical[1]	**0.065%**	0.065	0.10	0.10	0.035
Major	**0.65%**	0.65	0.65	0.65	0.83
Minor	**2.5/4.0%**	2.5	4.0	2.5	2.9
Particles	**0.65%**	ND	ND	ND	ND

[1] For Critical defects cited the most frequent response other than "zero".

ND = No Data, question not asked in survey from this year

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5.4 Do you have Alert / Action limits for 100% inspection results and what is done when the limit is exceeded?

Total respondents: 133

	2023	2014	2008	2003	1996
Firms with Limits	**92%**	89%	85%	76%	85%
Same for all Products	**54%**	40%	44%	32%	82%
Practice if Limit exceeded:					
Investigate	**91%**	88%	70%	95%	80%
Reinspect	**68%**	69%	45%	50%	82%
Reject	**28%**	29%	5%	36%	45%
Change Sampling Plan	**27%**	26%	ND	ND	ND
Other[1]	**6%**	ND	ND	ND	ND

[1] Actions vary depending on defect type

ND = No Data, question not asked in survey from this year

5.5 What are typical values used for Alert / Action limits for 100% inspection results?

Total respondents: 120

	2023	2014	2008	2003	1996
<1%	**21%**	19%	32%	29%	14%
1 to 2%	**23%**	36%	21%	41%	18%
3 to 5%	**32%**	29%	37%	29%	27%
5 to 10%	**14%**	14%	16%	35%	18%
>10%	**10%**	3%	2%	11%	9%

5.6 Is there a limit to the number of times a lot may be reinspected?

Total respondents: 126

	2023	2014	2008	2003	1996
Have a Limit	**ND**	99%	63%	ND	ND
Typical Limit used:					
Never	**2%**	ND	ND	ND	ND
1	**47%**	58%	40%	ND	ND
2	**41%**	35%	0%	ND	ND
3	**7%**	6%	60%	ND	ND
>3	**2%**	1%	ND	ND	ND

ND = No Data, question not asked in survey from this year

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5.7 Do you reinspect and return containers that are found to be acceptable after being culled out or rejected during initial inspection? This is often referred to as a two-stage inspection.

Total respondents: 116

	2023	2014	2008	2003	1996
After Manual Inspection	**ND**	**13%**	25%	22%	45%
After Automated Inspection	**60%**	**48%**	55%	38%	58%

ND = No Data, question not asked in survey from this year

5.8 What inspection method do you use for the second stage?

Total respondents: 66

- **Manual: 65%**
- Semiautomated: 17%
- Automated: 18%

5.9 Do you perform a separate and independent AQL of the reinspected part of lot before combining with the part accepted in the primary inspection?

Total respondents: 68

- **Yes: 77%**
- No: 23%

5.10 Does your firm use the same inspection methods and acceptance criteria for …?

Total respondents: 121

	2023	2014	2008	2003	1996
Different Geographic Regions	**88%**	78%	67%	87%	90%
Veterinary (vs. Human) Products	**89%**	77%	100%	83%	100%
Clinical (vs. Commercial)	**96%**	86%	ND	ND	ND

5.11 Special inspection situations and difficult to inspect products (DIP).

Total respondents: 79

	2023	2014	2008	2003	1996
Inspect empty containers	**23%**	15%	16%	28%	30%
For firms with applicable difficult to inspect products:					
Reconstitute lyo/powder	**67%**	86%	16%	28%	30%
Insp. after filling/before lyo	**11%**	14%	ND	ND	ND
Transfer to clear container	**14%**	14%	ND	ND	ND
Increase light intensity	**8%**	ND	ND	ND	ND

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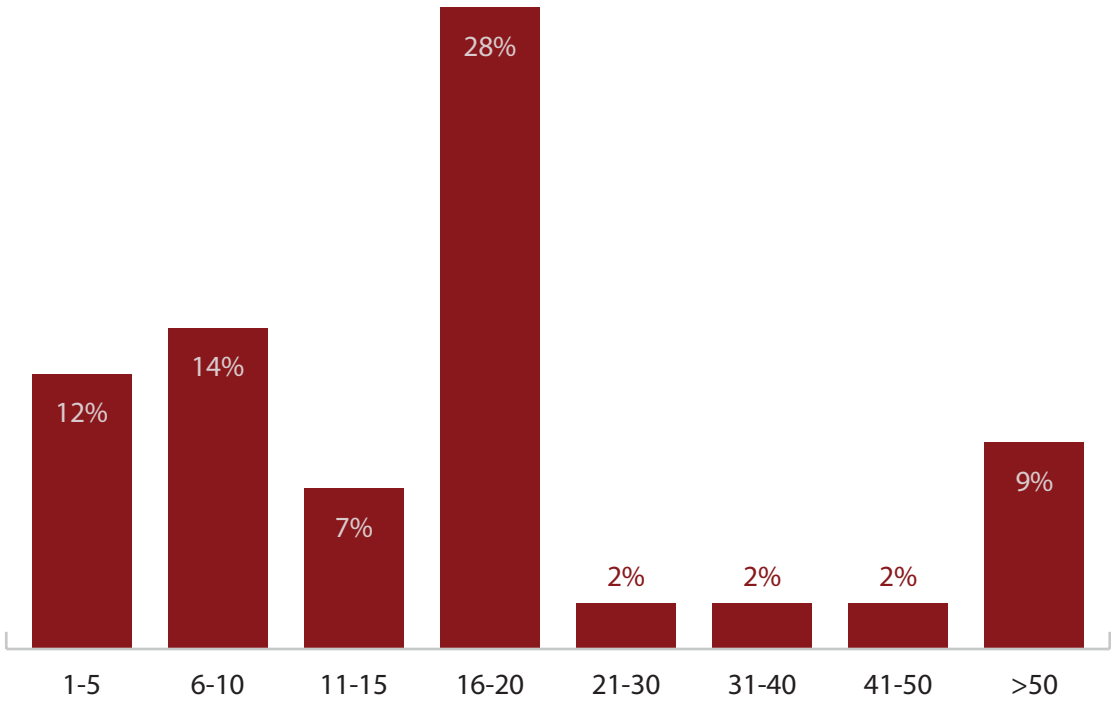
5.12 If you perform destructive testing on difficult to inspect products and/or containers, what is the typical sample size?

Total respondents: 76



Future Direction (Question 6.1 – 6.4)

6.1 31% of respondents expect to see changes in customer expectations for container and closure or visible particulate quality in the next five years.

Total respondents: 131

Summary of responses:

- They expect customer expectations to become "more stringent" and to receive "increased attention and stronger criteria".
- Multiple respondents indicate customer expectations in the future to "Move to 100% nondestructive CCI".

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6.2 40% of respondents have been challenged on their visual inspection procedures or acceptance criteria by a regulatory agency in the last two years.

Total respondents: 132

Summary of responses:

- Many responses indicate challenges regarding "kits used for the qualification of inspectors" with a focus on "composition of defect test sets, generation of defect test sets" and "size of test kits, bracketing approach, handling of test kits."
- Multiple responses indicated challenges on qualification approach for manual and automated inspection including "inspectors not challenged in fatigue condition" and "challenged only requiring 80% detection of major defects during operator qualification".
- A number of respondents noted challenges regarding "institute documented breaks for inspectors".

6.3 44% of respondents expect changes in the regulatory expectations for container and closure or visible particulate quality in the next five years.

Total respondents: 129

Summary of responses:

- They expect regulatory expectations to be "more stringent" or "with tighter limits".
- They see regulators pushing firms "to automate as soon as possible".
- They see increasing expectations for "a quantitative method for container closure integrity" testing per the update to Annex 1.
- They also see increased expectations to follow recommendations in USP <1790> including "destructive testing" of difficult to inspect products and a "life cycle approach" for defect reduction.

6.4 What other changes do you see in your inspection activities for injectable products in the next five years?

Summary of responses:

- Majority of responses indicated a move to automated inspection, the use of AI/DL, or implementing additional digital resources in the process.
- Several responses indicate tighter regulation may require 100% CCIT or automated inspection in the near future.
- Moving away from semi-automated inspection to automated or manual inspection.
- Continued focus on particle reduction, including the reduction of particle size when qualifying the inspection method, zero defect products, the use of calibrated particles, and continued particle control and monitoring programs.

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PDA Research: *2023 PDA Visual Inspection Survey*

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